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Thermodynamic optimization of the germanium–yttrium system

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Abstract The thermodynamic modelling of the germanium–yttrium binary system was carried out with the help of CALculation of PHAse Diagram method. Ge_3Y_5, Ge_4Y_5, $Ge_{10}Y_{11}$, GeY, $Ge_{1.5}Y$, Ge_5Y_3, Ge_2Y, and Ge_7Y_2 have been treated as stoichiometric compounds while a solution model has been used for the description of the liquid, BCC_A2 and HCP_A3 phases. The calculations based on the thermodynamic modelling are in good agreement with the phase diagram data and experimental thermodynamic values available in the literature.

Keywords Thermodynamic optimization of Ge–Y system · Calphad method · Phase diagram

Introduction

Germanium and alloys on its basis have super- and semiconducting properties and can be used as resistive materials. A deep knowledge of the thermodynamic properties of germanium alloys is needed to refine the Ge–Me phase diagrams and improve germanium alloy production methods.

The Ge–Y experimental phase diagram was determined by Schmidt et al. [1] using thermal, microscopic, chemical, and X-ray analyses. Nine stoichiometric intermetallic compounds were described. For the Ge_3Y_5 compound, a congruent melting was determined at 2238 K; for the Ge_4Y_5, $Ge_{10}Y_{11}$, GeY, $Ge_{1.5}Y$-LT, $Ge_{1.5}Y$-HT, Ge_5Y_3, Ge_2Y and Ge_7Y_2 compounds, a peritectic formation was reported, respectively, at 2218, 2173, 2003, 1733, 1183, 1153, and 1133 K see Fig. 1. A polymorphic transition ($Ge_{1.5}Y$_LT ⇔ $Ge_{1.5}Y$_HT) was determined at 1703 K for $Ge_{1.5}Y$ in [1]. According to Schmidt et al. [1], none of the intermediate phases showed a narrow homogeneity range. Consequently, they are drawn as stoichiometric in the assessed diagram. The maximum solubility of Y in (Ge) is <0.5 at% at the eutectic temperature (1093 K) and almost zero at 873 K. The maximum solubility of Ge in (Y) is below 1.0 at% at the eutectic temperature (1553 K) and ~0.3 at% at 1073 K. The crystal structures of the various phases of the Ge–Y system are reported in Table 1.

No thermodynamic description of the Ge–Y system is available in the literature. In this article, a thermodynamic optimization for the Ge–Y binary system is performed based on the experimental phase diagram data in the literature (experimental thermodynamic and phase diagram data).

Thermodynamic data

A number of thermodynamic examinations have been performed by high-temperature calorimetry only for $0.0 < x(Y) < 0.46$ [2–5] and for $0.8 < x(Y) < 1.0$ [3–5]. Nikolaenko and co-authors have measured the $\Delta_{mix}\bar{H}_{Ge}$ for $0.0 < x(Y) < 0.235$ at 1523 K [2] and for $0.0 < x(Y) < 0.423$ at 1900 K [3, 4]. The $\Delta_{mix}\bar{H}_{Ge}$ was also measured in [3, 4] for $0.0 < x(Ge) < 0.186$ at 1900 K. Esin et al. [5] have measured the $\Delta_{mix}\bar{H}_{Ge}$ for $0.0 < x(Y) < 0.46$ and the $\Delta_{mix}\bar{H}_{Ge}$ for $0.0 < x(Ge) < 0.2$ at 1920 K [5]. The data of

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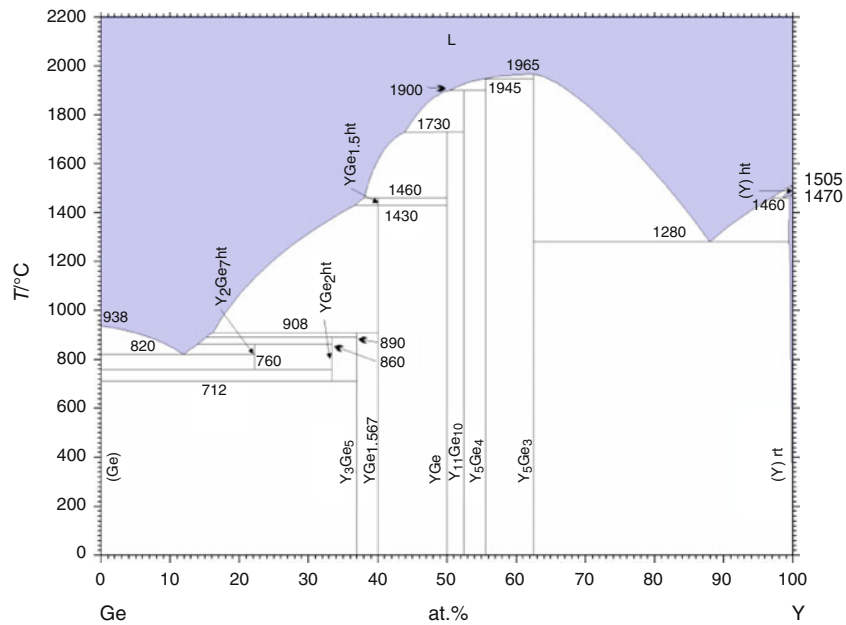

Fig. 1 The Ge–Y phase diagram drawn by Schmidt et al. [1]

Table 1 Symbols and crystal structures of the solid phases in the Ge–Y alloys [1]

Diagram symbol	Composition/ at% Y	Symbol used in thermo-calc data file	Space group	Prototype
(Ge)	0	DIAMOND_ FCC_A4	$Fd\bar{3}m$	C (diamond)
Ge_3Y_5	62.5	Ge_3Y_5	$P6_3/mcm$	Mn_5Si_3
Ge_4Y_5	55.55	Ge_4Y_5	$Pnma$	Sm_5Ge_4
$Ge_{10}Y_{11}$	52.32	$Ge_{10}Y_{11}$	$I4/mmm$	$Ho_{11}Ge_{10}$
GeY	50	GeY	$Cmcm$	CrB
$Ge_{1.5}Y$	40	$Ge_{1.5}Y$	$P6/mmm$	AlB_2
Ge_5Y_3	37.5	Ge_5Y_3	$I4_1/amd$	$ThSi_2$
Ge_2Y	33.33	Ge_2Y	–	–
Ge_7Y_2	22.22	Ge_7Y_2	$C222_1$	–
βY	100	BCC_A2	$Im\bar{3}m$	W
αY	100	HCP_A3	$P6_3/mmc$	Mg

Esin et al. [5] are significantly less negative than those of Nikolaenko et al. [2–4]. Moreover, the $\Delta_{mix}\bar{H}_{Ge}$ data reported in [3, 4] for $T = 1900$ K are different. Kanibolotsky et al. [6] measured the partial molar enthalpies of mixing of the liquid phase using a new high-temperature mixing calorimeter. These authors also calculated the integral molar enthalpies of mixing of the liquid phase at $T = 1770$ K.

Recently, Sudavtsova et al. [7] measured the partial molar enthalpies of mixing of the liquid phase using high-temperature isoperibolic calorimetry at 1780 K ± 5 over a concentration range $0 < x(Y) < 0.4$.

The enthalpies of formation of the intermetallic compounds of the Ge–Y system have been determined by several authors. Polotskaya et al. [8] measured the enthalpy of formation of Ge_3Y_5, Ge_4Y_5, $Ge_{10}Y_{11}$, GeY, and $Ge_{1.5}Y$ by EMF. The authors in Refs. [9–11] measured the enthalpy of formation of Ge_3Y_5 using calorimetry method. Miedema [12] predicted the enthalpy of formation of GeY based on a semi-empirical model.

Thermodynamic models

The Gibbs energy function ${}^0G_i^{\phi}(T) = G_i^{\phi}(T) - H_i^{SER}(298.15\,K)$ for the element i (i = Ge, Y) in the ϕ phase (ϕ = Liquid, DIAMOND_FCC_A4, BCC_A2 or HPC_A3) is described by an equation of the following form:

$$G_i^{\phi}(T) = a + bT + cT\ln T + dT^2 + eT^3 + fT^7 + gT^{-1} + hT^{-9}. \quad (1)$$

In this article, the Gibbs energy functions are taken from the SGTE compilation of Dinsdale [13]. These functions are valid up to $T = 3700$ and 3200 K for yttrium and germanium.

The solution phases (ϕ = Liquid, BCC_A2 and HPC_A3) are treated as a substitutional solution with the Gibbs energy expressed as:

$$G^{\phi} - H^{SER} = {}^{ref}G^{\phi} + {}^{id}G^{\phi} + {}^{ex}G^{\phi} \quad (2)$$

$${}^{ref}G_{liq} = x_{Ge}\left[{}^0G_{Ge}^{liq} - H_{Ge}^{SER}\right] + x_Y\left[{}^0G_Y^{liq} - H_Y^{SER}\right], \quad (3)$$

where H_i^{SER} (298.15 K) is the molar enthalpy of the i element at 298.15 K in its standard element reference

(SER) state, DIAMOND_FCC_A4 for Ge and HCP_A3 for Y.

$$^{id}G^{\phi} = RT(y_{Ge}\ln y_{Ge} + y_{Y}\ln y_{Y}), \quad (4)$$

where R is the ideal gas constant, T is the temperature, in Kelvin; x_{Ge} and x_{Y} are the mole fractions of the elements Ge and Y, respectively.

The $^{ex}G^{\phi}$ energy part in Eq. 2 is given by the Redlich–Kister polynomial [14]

$$^{ex}G^{\phi} = y_i y_j \sum_{\lambda=0}^{\lambda} {}^{\lambda}L^{\phi}_{i,j}(y_i - y_j)^{\lambda} \quad (5)$$

with i and j are indices which correspond to the two species germanium and yttrium.

The binary interaction parameters of the $^{\lambda}L^{\phi}_{i,j}$ type, assessed in this study, were temperature dependent as follows:

$$^{\lambda}L^{\phi}_{i,j} = a_{\lambda} + b_{\lambda}T \quad (6)$$

a_{λ} and b_{λ} are the coefficients to be optimized.

Nine intermetallic phases Ge_3Y_5, Ge_4Y_5, $Ge_{10}Y_{11}$, GeY, $Ge_{1.5}$Y-LT, $Ge_{1.5}$Y-HT, Ge_5Y_3, Ge_2Y, and Ge_7Y_2 were considered as stoichiometric compounds, their Gibbs energy noted as $^{0}G_{ApBq}$ was expressed as follows:

$$^{0}G_{ApBq} = \frac{p}{p+q}{}^{0}G_{A} + \frac{q}{p+q}{}^{0}G_{B} + a + bT, \quad (7)$$

where $^{0}G_{A}$ and $^{0}G_{B}$ are the Gibbs energy of the pure elements germanium and yttrium, respectively; a and b are parameters to be determined.

Results and discussions

The parameters of the thermodynamic modelling of the Ge–Y binary system were optimized using the Parrot module [15] in ThermoCalc [16] with the CALPHAD [17] technique. This program is able to incorporate the contributions of different types of experimental data in one operation by simultaneously minimizing the sums of the error for each selected piece of data with a given weight. The chosen weight is based upon its relative importance, experimental uncertainty, and the validity of the data as determined by the judgment of the modeller. All calculations in this study are performed using the ThermoCalc software. The reference state of the Gibbs energy of the individual phases is the SER, i.e., the enthalpies of the pure elements in their stable states at 298.15 K.

Most of the experimental data which are mentioned in "Thermodynamic data" are used in this study. In this parameter optimization procedure, for modeling the liquid phase we first imposed the conditions $d^2G/dx^2 > 0$ (additional constraints to avoid the appearance of an unwanted inverted miscibility gap in the liquid phase during the phase diagram calculation as recommended in [18–20]). The phase boundary data reported by Schmidt et al. [1] and thermodynamic data (the integral and the partial molar enthalpies of mixing of the liquid phase mentioned above) were used. We then modelled the parameters of the intermetallic phases beginning with the congruent melting (Liq ⇔ Ge_3Y_5).

Table 2 The optimized thermodynamic parameters of the Ge–Y system

Phase	Thermodynamic model	Parameters units in SI
Liquid	(Ge, Y)	$^{0}L^{liq}_{Ge,Y} = -325857.633 + 72.4738522T + 6.51248164.10^{-13}\ Tln(T)$
		$^{1}L^{liq}_{Ge,Y} = 16510.5176 - 4.65893352\ T$
		$^{2}L^{liq}_{Ge,Y} = 43797.7362 - 0.090413928\ T$
BCC_A2	$(Ge, Y)_1(Va)_3$	$^{0}L^{BCC_A2}_{Ge,Y} = -141713.309$
HCP_A3	$(Ge, Y)_1\ (Va)_{0.5}$	$^{0}L^{HCP_A3}_{Ge,Y} = -19680.9005$
DIAMOND_FCC_A4	(Ge)	No excess term
Ge_7Y_2	$(G)_{0.778}$:$(Y)_{0.222}$	$G^{Ge7Y2}_{Ge:Y} - 0.778\,^{298}H^{DIAMOND_FCC_A4}_{Ge} - 0.222\,^{298}H^{HCP}_{Y} = -42349 + 0.64\ T$
Ge_2Y	$(Ge)_{0.667}$:$(Y)_{0.333}$	$G^{Ge2Y}_{Ge:Y} - 0.667\,^{298}H^{DIAMOND_FCC_A4}_{Ge} - 0.333\,^{298}H^{HCP}_{Y} = -69577 + 6.83\ T$
Ge_5Y_3	$(Ge)_{0.375}$:$(Y)_{0.625}$	$G^{Ge3Y5}_{Ge:Y} - 0.375\,^{298}H^{DIAMOND_FCC_A4}_{Ge} - 0.625\,^{298}H^{HCP}_{Y} = -79833.8303 + 9.2\ T$
$Ge_{1.5}$Y-LT	$(Ge)_{0.6}$:$(Y)_{0.4}$	$G^{Ge1.5Y-LT}_{Ge:Y} - 0.6\,^{298}H^{DIAMOND_FCC_A4}_{Ge} - 0.4\,^{298}H^{HCP}_{Y} = -84785.4205 + 9.679\ T$
$Ge_{1.5}$Y-HT	$(Ge)_{0.6}$:$(Y)_{0.4}$	$G^{Ge1.5Y-HT}_{Ge:Y} - 0.6\,^{298}H^{DIAMOND_FCC_A4}_{Ge} - 0.4\,^{298}H^{HCP}_{Y} = -83024\ .9 + 8.64788944\ T$
GeY	$(Ge)_{0.5}$:$(Y)_{0.5}$	$G^{GeY}_{Ge:Y} - 0.5\,^{298}H^{DIAMOND_FCC_A4}_{Ge} - 0.5\,^{298}H^{HCP}_{Y} = -101529\ .617 + 14.5119619\ T$
$Ge_{10}Y_{11}$	$(Ge)_{0.477}$:$(Y)_{0.523}$	$G^{Ge10Y11}_{Ge:Y} - 0.477\,^{298}H^{DIAMOND_FCC_A4}_{Ge} - 0.523\,^{298}H^{HCP}_{Y} = -104815\ .363 + 15.7630509\ T$
Ge_4Y_5	$(Ge)_{0.445}$:$(Y)_{0.555}$	$G^{Ge4Y5}_{Ge:Y} - 0.445\,^{298}H^{DIAMOND_FCC_A4}_{Ge} - 0.555\,^{298}H^{HCP}_{Y} = -109546.51 + 18.117368\ T$
Ge_3Y_5	$(Ge)_{0.375}$:$(Y)_{0.625}$	$G^{Ge3Y5}_{Ge:Y} - 0.375\,^{298}H^{DIAMOND_FCC_A4}_{Ge} - 0.625\,^{298}H^{HCP}_{Y} = -107409 + 19.01665\ T$

Va vacancy

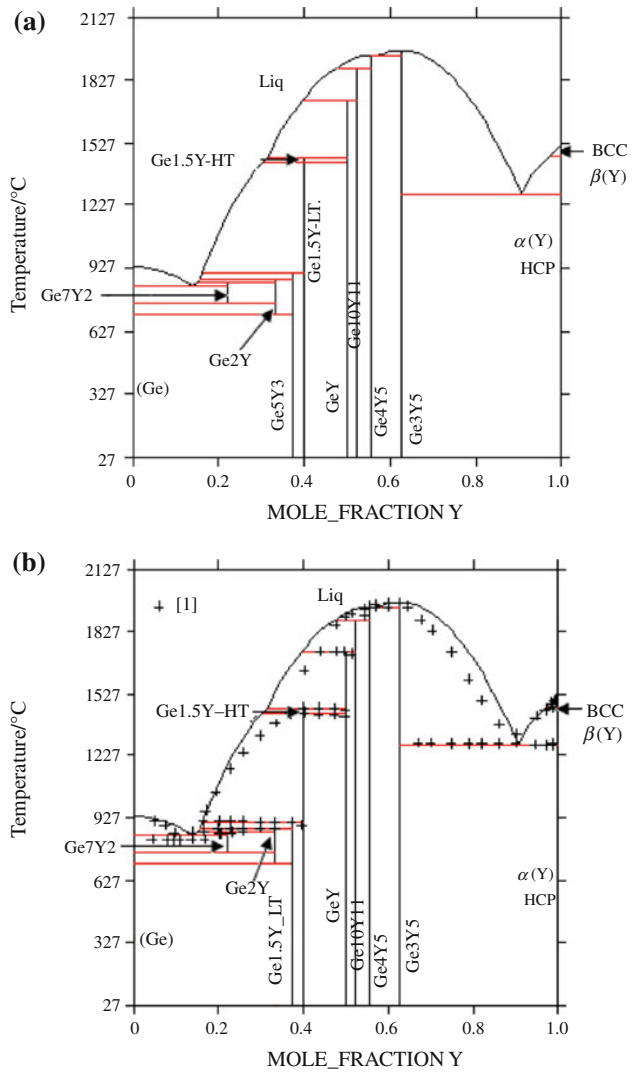

Fig. 2 **a**, **b** Calculated Ge–Y phase diagram with the experimental data

Finally, all the parameters of the different phases were simultaneously optimized with the experimental data mentioned above. All the parameters are evaluated and listed in Table 2.

The phase diagram calculated by the present parameters is shown in Fig. 2a. The calculated results are compared with the experimental data in Fig. 2b and are nearly identical to the ones represented by Schmidt et al. [1]. The calculated invariant reactions are listed in Table 3 and compared with the experimental results from [1]. The temperature of invariant reactions fits well the experiments [1] except both the temperature of the invariant reaction: (Ge) + Ge_7Y_2 ⇔ Liq (calculated 1113 K, measured 1093 K) and $Ge_{10}Y_{11}$ ⇔ Liq + Ge_4Y_5 (calculated 2157 K, measured 2173 K).

The calculated integral enthalpy of mixing of the liquid Ge–Y alloys referring to liquid components at the temperatures range (820 K < T < 1920 K) is plotted in Fig. 3. Also shown in this figure are the experimental data. The assessed integral enthalpies of liquid agree with the experimental data. In Fig. 4 are shown the calculated partial enthalpies of mixing of germanium and yttrium of the liquid phase, compared with the ones measured by the authors in Ref. [2–6]. A satisfactory agreement is observed with the experimental data in Ge-rich and Y-rich regions.

Table 3 Invariant reactions in the Ge–Y system

Reactions	[1]		This study	
	T/K	X_{Liq}/ at%Y	T/K	X_{Liq}/ at%Y
(Ge) + Ge_7Y_2 ⇔ Liq	1093	0.119	1113	0.137
(Ge) + Ge_5Y_3 ⇔ Ge_2Y	985	–	982	–
Ge_7Y_2 ⇔ Ge_2Y + (Ge)	1033	–	1031	–
Ge_7Y_2 ⇔ Ge_2Y + Liq	1133	0.142	1132	0.150
Ge_2Y ⇔ Liq + Ge_5Y_3	1163	0.152	1150	0.155
Ge_5Y_3 ⇔ Liq + $Ge_{1.5}Y$-LT	1181	0.164	1177	0.161
$Ge_{1.5}Y$-HT ⇔ GeY + Liq	1732	0.384	1733	0.314
GeY ⇔ Liq + $Ge_{10}Y_{11}$	2003	0.44	2003	0.396
$Ge_{10}Y_{11}$ ⇔ Liq + Ge_4Y_5	2173	0.498	2157	0.477
Ge_4Y_5 ⇔ Liq + Ge_3Y_5	2218	0.552	2214	0.572
Ge_3Y_5 ⇔ Liq	2238	0.625	2241	0.625
Liq ⇔ HCP + Ge_3Y_5	1553	0.88	1550	0.905
HCP ⇔ BCC + Liq	1733	0.97	1736	0.973
$Ge_{1.5}Y$-HT + Liq ⇔ $Ge_{1.5}Y$-LT	1703	0.366	1707	0.491
$Ge_{1.5}Y$-HT + GeY ⇔ $Ge_{1.5}Y$-LT	1703	0.384	1707	0.303

The enthalpies of formation of the Ge–Y intermetallic compounds measured and calculated ones are collected in Table 4 and also presented in Fig. 5. The optimized enthalpy of formation of Ge_3Y_5 ($\Delta H^f = -107.41$ kJ/mol) is somewhat slightly higher than the one determined in [9, 11] ($\Delta H^f = -89.8$ kJ/mol), the enthalpy measured by the authors in Ref. [10] is in reasonable agreement with our calculation.

The enthalpies of formation of the two polytypes: $Ge_{1,5}Y$_LT and $Ge_{1,5}Y$_HT are very similar and confirm the conclusion by Stein et al. [21], concerning the small structural energies of the different Laves phase polytypes. As $Ge_{1,5}Y$_LT is stable at low temperature, its enthalpy is a very little bit higher than the one of $Ge_{1,5}Y$_HT. On the contrary, its entropy is a little bit lower than the one of $Ge_{1,5}Y$_HT. It is necessary to use all the digits of the optimized parameters listed in Table 3 for the calculation of the polymorphic transition temperature between the polytypes at $T = 1710$ K.

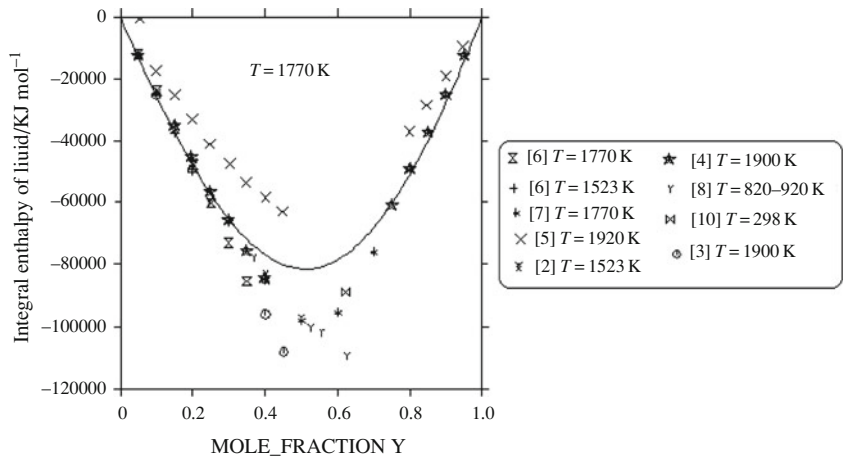

Fig. 3 Comparison of the calculated and measured integral enthalpies of mixing for the liquid phase

As first stated by [21], we verified that when the liquid phase is suspended, the intermetallic compounds are no more stable at high temperature and the calculated phase diagram gives the only BCC_A2 solid solution calculated on the whole yttrium composition range and not the DIAMOND_FCC_A4 one in the Ge-rich region (Fig. 6). This is due to the power series in terms of temperature for the Ge element in the BCC_A2 state which becomes metastable compared to the DIAMOND_FCC_A4 state around 1500 K and for higher temperatures [13].

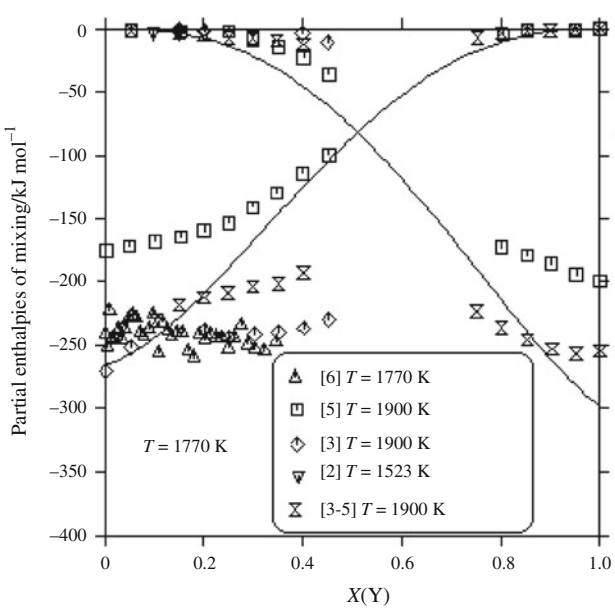

Fig. 4 Calculated yttrium partial enthalpies of mixing with the experimental data

Table 4 The enthalpies of formation of the Ge–Y intermetallic alloys

Compound	ΔH^f/kJ mol^{-1}	Method	References
Ge_7Y_2	−42.349	Optimization $T = 298.15$ K	This study
Ge_2Y	−69.577	Optimization $T = 298.15$ K	This study
Ge_5Y_3	−79.834	Optimization $T = 298.15$ K	This study
$Ge_{1.5}Y$-LT	−77.8 ± 3.4	EMF 820–920 K	[8]
	−69.664	Optimization $T = 870$ K	This study
	−84.785	Optimization $T = 298$ K	This study
$Ge_{1.5}Y$-HT	−84.1 ± 3.7	EMF 820–920 K	[8]
	−67.903	Optimization $T = 870$ K	This study
	−83.025	Optimization $T = 2985$ K	
GeY	−98 ± 4.9	EMF 820–920 K	[8]
	−108.8	Calculation	[12]
	−86.218	Optimization $T = 870$ K	This study
	−101.53	Optimization $T = 298$ K	This study
$Ge_{10}Y_{11}$	−100.5 ± 5.2	EMF 820–920 K	[8]
	−89.461	Optimization $T = 870$ K	This study
	−104.82	Optimization $T = 298$ K	This study
Ge_4Y_5	−102.5 ± 5.5	EMF 820–920 K	[8]
	−94.131	Optimization $T = 870$ K	This study
	−109.55	Optimization $T = 298$ K	This study
Ge_3Y_5	−109.7 ± 5.7	Calorimetry $T = 298.15$ K	[10]
	−89.8 ± 5		[9, 11]
	−91.861	Calorimetry $T = 298.15$ K	This study
	−107.41	Optimization $T = 870$ K	This study
		Optimization $T = 298$ K	

State element reference: $T = 298.15$ K, Ge diamond_A4 $T = 298.15$, P = 1e5 and Y hcp_A3 $T = 298.15$, P = 1e5

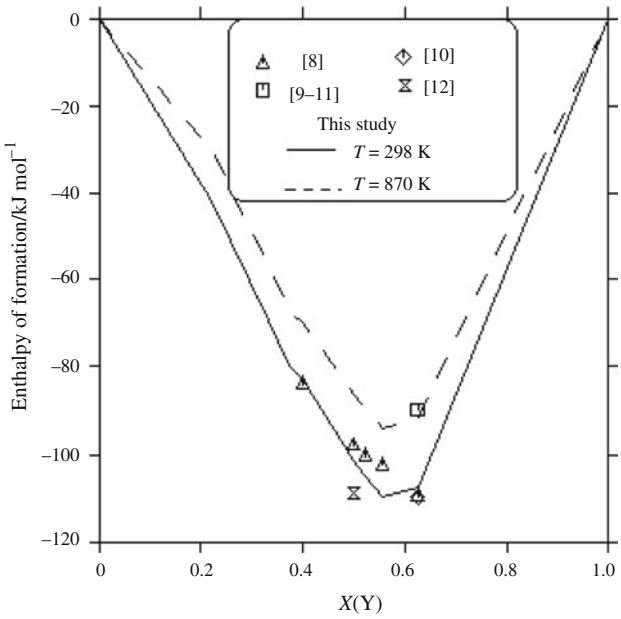

Fig. 5 Calculated and measured enthalpies of formation of the intermetallic compounds

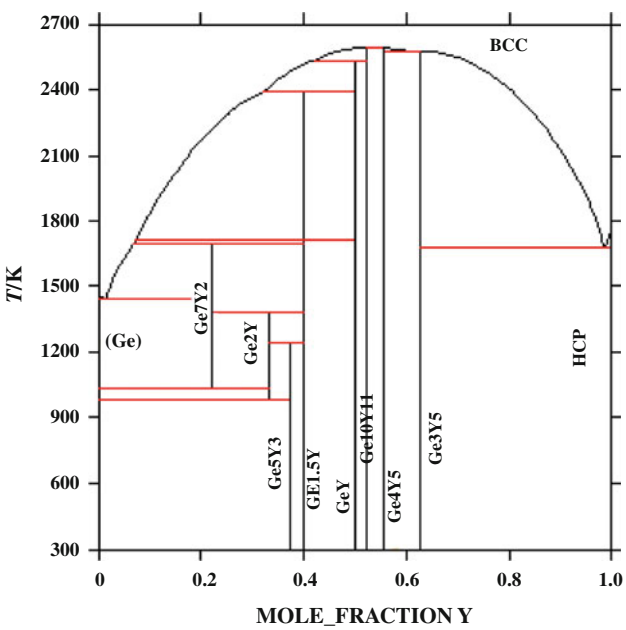

Fig. 6 Calculated Ge–Y phase diagram when the liquid phase is suspended

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